



CALICUT UNIVERSITY – FOUR-YEAR UNDER GRADUATE PROGRAMME (CU-FYUGP)

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	INTRODUCTORY MEDICAL PHYSICS				
Type of Course	Major Elective (SPECIALIZATION III: PHYSICS IN BIOLOGY)				
Semester	V				
Academic Level	300 - 399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	A strong foundation in physics, mathematics, and basic biology concepts.				
Course Summary	The medical physics course provides an interdisciplinary exploration of the application of physics principles to medical imaging, radiation therapy, and radiation protection, emphasizing the theoretical and practical aspects essential for understanding and contributing to advancements in medical diagnostics and treatment.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the concept of biometrics and its	Understanding	Conceptual Knowledge	Written exams, quizzes
CO2	Analyze bioelectric potentials and their	Analyzing	Conceptual Knowledge	Laboratory reports, projects
CO3	Identify and explain the major physiological	Understanding	Conceptual Knowledge	Presentations, written exams
CO4	Describe the principles underlying medical	Understanding	Conceptual Knowledge	Practical assessments, exams

CO5	Apply the principles of medical imaging to	Applying	Procedural Knowledge	Case studies, laboratory work
CO6	Evaluate the cognitive and technical aspects	Evaluating	Metacognitive Knowledge	Oral exams, practical exams
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Marks (70)
I	Biometrics – Man as a Physical Instrument		9	13
	1	Features of biomedical instrumentation system: Range, sensitivity, linearity, hysteresis, frequency response, Accuracy, Signal to noise ratio, stability, simplicity.	2	
	2	Aspects of Man-instrument system: Information gathering, diagnosis, Evaluative, Monitoring, Control.	2	
	3	Components of the Man-Instrument system: The subject, stimulus, the transducer, signal conditioning equipment, display equipment, Recorder.	2	
	4	Physiological systems of the body - Biochemical system, cardio-vascular, respiratory, nervous systems. (Sufficient exercises)	3	
	Sections 1.2-1.6 of Book 1			
II	Bioelectric Potentials and Major Physiological Systems of Human Body		22	32
	5	Sources of Bioelectric potentials: Resting and action potentials, propagation of action potentials	2	
	6	Bio-electric potentials, ECG, EEG, EMG.	2	
	7	The heart and cardiovascular system: The heart, blood pressure, characteristics of blood flow	3	
	8	Electrocardiography-electrodes and leads, principles of recording, Measurement of blood pressure, direct and indirect methods (two methods, qualitative ideas only).	3	
	9	Measurements in respiratory system: The physiology of respiratory system	3	

	10	Mechanics of breathing-working of Spirometer.	1	
	11	Nervous system-The anatomy of nervous system, neuronal communication	3	
	12	Measurements from the nervous system, neuronal firing measurements	2	
	13	Principles of EEG and EMG.	3	
	Sections 3.1-3.3, 5.1-5.4, 6.1, 6.2, 8.1, 8.2, 10.1, 10.2, 10.7 of Book 2			
III	Principles of Medical Imaging -1		8	12
	14	Ultrasonic Imaging: properties of ultrasound	2	
	15	modes of ultrasound transmission-pulsed, continuous, pulsed Doppler, ultrasound imaging, ultrasonic diagnosis, ultrasonic transducers	2	
	16	Generation of Ionizing Radiation	2	
	17	Instrumentation for Diagnostic X Rays	2	
	18	Special Techniques	2	
	Sections 9.2, 9.3, 14.1-14.3 of Book 1			
IV	Principles of Medical Imaging-2		9	13
	19	Radio-isotopes in Medical Diagnosis, Physics of Radioactivity	2	
	20	The Gamma Camera, Emission Computed Tomography (ECT), Positron Emission Tomography (PET Scanner)	2	
	21	Principles of NMR Imaging Systems, Image Reconstruction Techniques, Basic NMR Components.	3	
	22	Biological Effects of NMR Imaging, Advantages of NMR Imaging System	2	
	Sections 21.1, 21.2, 21.7, 21.9, 21.11, 22.1-22.5 of Book 2			
V	Open Ended Module – Suggestive topics		12	
	Biological effects of radiation, In vitro and in vivo testing, gamma rays for imaging, radio pharmaceuticals, the gamma camera, single photo emission computed tomography (SPECT), typical nuclear medicine images and normal and abnormal manifestations (<i>Techniques for radiation dosimetry by K Mahesh and D R Vij, Wiley Eastern Limited</i>) Lasers in Medicine - effects of laser radiation on tissue, surgical uses of Lasers, ophthalmic uses, photodynamic therapy, laser hazards-biological effects, photo thermal effects, photochemical effects, laser hazards to the			

	eye, to skin, safe exposure. (<i>Lasers in Medicine - An Introductory Guide, Gregory Absten, Springer Science Publications</i>)		
Books and References: 1. Biomedical Instrumentation and measurement”, Leslie Cromwell, Prentice Hall of India, New Delhi (Book 1) 2. Biomedical Instrumentation by R S Khandpur, Tata Mc Graw Hill Publication, New Delhi (Book 2)			

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO 4	PS O5	PS O6	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7
CO 1	0	1	0	0	0	0	2	2	2	0	0	0	0
CO 2	0	3	1	0	0	0	2	2	2	0	0	0	0
CO 3	0	0	2	0	0	0	2	2	2	0	0	0	0
CO 4	0	0	0	2	0	0	2	2	2	0	0	0	0
CO 5	0	0	0	0	2	0	2	2	2	0	0	0	0
CO 6	0	0	0	0	0	3	2	2	2	0	0	0	0

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments /Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory /Practical Exam	Assignment /Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6	✓	✓		✓